

and governments, which in turn can impact their investment decisions and spending.

12.3 MEANING OF BONDS AND THEIR CLASSIFICATION

12.3.1 Meaning of Bonds

A bond is a long-term debt instrument in which the issuer of the bond is bound to pay a fixed or floating interest rate and repay the principal amount after a fixed period. Thus, a bond is a way of taking long-term funds.

A Real-Life Example: On March 14, 2023, Tata Capital Financial Services Limited (TCFSL) issued **bonds** worth ₹ 2,000 crore with a coupon rate of 8.3% p.a. and a maturity period of 2 years and 11 months. It means that TCFSL will receive ₹ 2,000 crore and pay fixed interest at 8.3% annually i.e., ₹ 16,600. These ₹ 2,000 crore will be repaid at the end of 2 years 11 months.

12.3.2 Characteristics of Bonds

- 1) **Long-term instruments** – They are long-term debt instruments.
- 2) **Par value** – The par value is the stated face value of the bond. This value is mentioned on the bond certificate. It does not change.
- 3) **Carry coupon** – Except in the case of zero-coupon bonds, other bonds carry a fixed or floating interest rate known as a coupon rate. Such interest, generally, accrues annually, but it may be semi-annually or quarterly too. Moreover, interest is calculated on par or face value.
- 4) **Maturity period** – Bonds may have an indefinite maturity period, but generally they have a definite maturity period.
- 5) **Issued at par, premium, or discount** – They may be issued at par, premium or discount.
 - a) If the issued price of a bond exceeds its face value, then it is said that the bond has been **issued at a premium**.
 - b) If the issued price is equal to its face value, then it is said that the bond has been **issued at par**.
 - c) If it is issued at a price less than its face value, then this is called a bond **issued at a discount**.
- 6) **Redemption** – Bonds may be redeemed at par or premium. If the redemption value of a bond exceeds its face value, then this is called redemption at a premium; if the redemption value of a bond is equal to its face value, then this is called redemption at par. Theoretically, redemption at a discount is also possible, but practically, it does not seem

to be feasible for an investor that he would get less than what he had invested.

12.3.3 Classification of Bonds

1) Bonds Based on Issuer

- a) **Government bonds** – As the term implies these are those bonds which are issued by the Government (Central or State governments in the case of India). Indian bond market is majorly dominated by Government bonds. These bonds generally offer stable returns and are considered very safe because they are guaranteed by the Indian government. The interest rate on G-sec (a short form for Government securities or bonds) varies between 7% and 10%. In India, Government bonds may take the following forms:
 - i) **Fixed-rate bonds (FRBs)** – These bonds yield fixed interest payments throughout the life of the bonds.
 - ii) **Floating rate bonds** – These bonds yield variable interest payments.
 - iii) **Sovereign Gold Bonds (SGBs)** – These are the schemes in which an investor is allowed to invest in digital gold at an average price of gold without carrying gold in physical form. The Sovereign Gold Bond (SGB) Scheme 2022-23 is an example in which the interest rate is 2.5% p.a. on the nominal value of the bonds and they are redeemable on expiry of 8 years.
 - iv) **Inflation Indexed Bonds** – These bonds have a unique feature i.e., their prices and interest payments are adjusted according to the price level or inflation rate based on the consumer price index (CPI) or wholesale price index (WPI).
 - v) **7.75% GOI Savings Bonds** – This government security was launched in 2018 to replace the 8% savings bond. The interest rate applicable here is 7.75%. The RBI stipulates that these bonds can be in the possession of individual(s) who aren't NRIs, minors, or are a Hindu undivided family.
- b) **Corporate bonds** – These bonds are issued by corporations or companies. Since, Tata Capital Financial Services is a company in the above example, therefore, the bonds issued by it are corporate bonds. These bonds are generally considered to be riskier than government bonds. The creditworthiness of the issuing company is an important factor in determining the risk associated with the bond, as well as the interest rate offered to investors. Companies with high credit ratings are generally considered to be more creditworthy and may offer lower interest rates, while companies with lower credit

ratings may offer higher interest rates to compensate for the higher risk of default. A few examples of corporate bonds are:

- i) *LIC Housing Finance Limited Bonds* (coupon rate 7.95%, maturity date January 29, 2028, issue size ₹ 1477 crores)
 - ii) *Hindustan Petroleum Bonds* (coupon rate 7.74%, maturity date March 2028, issue size
 - iii) *Indian Railway Finance Corporation Limited Bonds* (issued date February 23, 2012, coupon rate 8.1%, maturity date February 23, 2027, face value ₹ 1,000).
- c) **Municipal bonds (or muni)** – These bonds are issued by local authorities or local governments. For example, on February 20, 2023, *Indore Municipal Corporation* allotted bonds with a coupon rate of 8.25%, maturity date of February 20, 2026, and issue size ₹ 61crore.
- d) **Foreign bonds** – These are bonds issued by a foreign government or foreign corporation.

2) Bonds Based on Coupon

- a) **Zero coupon bonds (or pure discount bonds)** – These bonds do not carry any interest or coupon, but are issued at a discount and repaid/redeemed at par. For example, suppose that a bond has a face value of ₹ 100 and this is issued at a discount of ₹ 10, therefore, the issuer receives ₹ 90. Moreover, it will be repaid at ₹ 100. Thus, an investor by paying ₹ 90 will get ₹ 100 on the maturity of the bond. Thus, in the case of zero-coupon bonds, an investor earns capital return only not revenue return in the form of interest.
- b) **Coupon bonds (or regular bonds)** – These bonds carry some interest. If they carry fixed interest, they are known as **fixed coupon bonds** and if they carry floating interest, they are known as **floating coupon bonds**. Thus, in the case of coupon bonds, an investor earns capital as well as revenue returns.

3) Bonds based on maturity

- a) **Perpetual bonds** – These bonds have indefinite maturity.
- b) **Non-perpetual bonds** – These bonds have a definite maturity period.

4) Bonds based on option

- a) **Puttable bond (or retractable bonds)** – A put bond is a type of bond that gives the bondholder the right to sell the bond back to the issuer at a predetermined price and a specified date before the

bond's maturity. The price at which the bond can be sold is called the put price, and it is usually higher than the bond's face value.

- b) **Callable bond** – A callable bond is a type of bond that gives the issuer the right to redeem the bond before its maturity date. This means that the issuer can call or buy back the bond from the bondholder before the scheduled maturity date at a specified call price.

5) Bonds based on convertibility

- a) **Convertible bonds** – These bonds can be converted into the shares of the issuer company depending upon the terms and conditions of the issue.
- b) **Non-convertible bonds** – These bonds cannot be converted into the shares of the issuer company but they, generally, carry a higher interest rate than convertible bonds.

Check Your Progress 1

- 1) What do you mean by bonds? What are their characteristics?

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- 2) What are the types of bonds?

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12.4 VALUATION OF BONDS

Valuation of bonds means finding the **intrinsic value of a bond**. This intrinsic value is simply known as **the value of a bond**. The value of a bond is nothing but the present value of all the coupons and the redemption value of the bond. Therefore, the value of a bond can be calculated using the following formula:

$$V = \left[\frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \dots + \frac{C_n}{(1+i)^n} \right] + \frac{RV}{(1+i)^n}$$

$$V = \left[C_1 \times \frac{1}{(1+i)^1} + C_2 \times \frac{1}{(1+i)^2} + \dots + C_n \times \frac{1}{(1+i)^n} \right] + RV \times \frac{1}{(1+i)^n}$$

$$V = \left[\sum_{k=1}^n C_k \times \frac{1}{(1+i)^k} \right] + \frac{RV}{(1+i)^n}$$

Where,

$C_1, C_2, C_3, \dots, C_n$ are coupons or interest payments over n years, RV is the redemption value, and i is the discount rate. Apart, from the expressions

$$\frac{1}{(1+i)^1}, \frac{1}{(1+i)^2}, \dots, \frac{1}{(1+i)^n}$$

are known as the **present value factors (PVFs)** calculated for the first year, second year, third year, and so on. These present value factors can be calculated using

- Present value tables,
- log technique,
- financial or scientific calculators,
- MS-Excel
- Various apps e.g., <https://cleartax.in/s/present-value-calculator>

Value of a Perpetual Bond

If a bond is a perpetual bond, then

$$V = \frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \dots + \frac{C_n}{(1+i)^n} + \dots \dots \dots$$

If the coupons are constant i.e., $C_1 = C_2 = C_3 = \dots = C$, then

$$V = \frac{C}{(1+i)^1} + \frac{C}{(1+i)^2} + \dots + \frac{C}{(1+i)^n} + \dots \dots \dots$$

This is obvious that the above series is an infinite geometric progression with a common ratio

$$\frac{1}{1+i}$$

We know that if a is the first term and r is the common ratio of an infinite geometric progression, then its sum is given by the following formula:

$$S_{\infty} = \frac{a}{1-r}$$

Therefore,

$$V = \frac{C/(1+i)}{1-1/(1+i)}$$

$$V = \frac{C/(1+i)}{(1+i-1)/(1+i)}$$

$$V = \frac{C}{i}$$

The decision to buy bond:

- 1) **Overvalued bond** – If the market price of the bond exceeds its value, then it is called an overvalued bond, therefore, it should not be purchased.
- 2) **Undervalued bond** – If the market price of the bond is less than its value, then it is called an undervalued bond, therefore, it should be purchased.

The decision to sell a bond

Suppose you are the holder of a bond and you find that the market price of the bond exceeds its value, then you should sell the bond in the market. If the market price of the bond is less than its value, then it is better to hold it.

Example 12.1: Suppose that a corporation issues a bond having a face value ₹ 1,000 redeemable at a premium of 10% at the end of 3 years. If the discount rate is 12% and the coupon rate is 8%, then calculate the value of this bond. Moreover, would you buy this bond if the market price of the bond is ₹ 1,050 or ₹ 800?

Solution

Step 1: Summarise the information

Face value = ₹ 1,000

Coupon rate = 8%

Coupon = Coupon rate × face value = 8% × ₹ 1,000 = ₹ 80

Discount rate = i = 12% = 0.12

Maturity of bond = n = 3 years

Redemption value = RV = Face value + premium of 10% = 1,000 + 10% × 1,000 = ₹ 1100

Step 2: Apply the following formula

$$V = \left[\frac{C_1}{(1+i)^1} + \frac{C_2}{(1+i)^2} + \frac{C_3}{(1+i)^3} + \dots + \frac{C_n}{(1+i)^n} \right] + \frac{RV}{(1+i)^n}$$

$$\begin{aligned}
&= \left[80 \times \frac{1}{(1+0.12)^1} + 80 \times \frac{1}{(1+0.12)^2} + 80 \times \frac{1}{(1+0.12)^3} + \right] + 1100 \times \frac{1}{(1+0.12)^3} \\
&= \left[80 \times \frac{1}{(1.12)^1} + 80 \times \frac{1}{(1.12)^2} + 80 \times \frac{1}{(1.12)^3} + \right] + 1,100 \times \frac{1}{(1+0.12)^3} \\
&= [80 \times 0.8929 + 80 \times 0.7972 + 80 \times 0.7118 +] + 1,100 \times 0.7118 \\
&= 71.43 + 63.76 + 56.94 + 782.98 \\
&= 975.13
\end{aligned}$$

Thus, the value of the bond is ₹ 975.13.

The decision to buy bond:

- 1) If the market price is ₹ 1,050, then it exceeds the value ₹ 975.13, therefore, the bond is overvalued and should not be bought.
- 2) If the market price is ₹ 800, then it is less than the value of ₹ 975.13, therefore, the bond is undervalued and should be bought.

Example 12.2: What is the value of a perpetual bond with a coupon of 8% and face value ₹ 1000? Assume a discount rate of 10%.

Solution

$$\text{Coupon} = C = 8\% \times ₹ 1,000 = ₹ 80$$

$$\text{Discount rate} = i = 10\% = 0.10$$

Therefore, the value of the bond is

$$V = \frac{C}{i} = \frac{80}{0.10} = ₹ 800$$

Check Your Progress 2

- 1) How are bonds valued?

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- 2) A bond with a face value of ₹ 5,000 matures at the end of 5 years, the rate of interest on the bond being 15% p.a. paid annually. Find the value of the bond to yield a return of 10% p.a. Further, if the bond is trading at ₹ 6,000, then is it worth buying it? Assume that the bond will be redeemed at par. (Hint: Find the present value of all the benefits from the bond. Refer to example 12.1)

- 3) Will your decision be different if the bond is traded at ₹ 5,500? (**Hint: Refer to example 12.1**)

12.5 BOND YIELDS

12.5.1 Yield to Maturity (YTM)

From the previous section, we have the formula to calculate the value of a bond i.e.,

$$V = \left[\sum_{k=1}^n C_k \times \frac{1}{(1+i)^k} \right] + \frac{RV}{(1+i)^n}$$

So, if we have the values of C_k for $k = 1, 2, 3, \dots, n$, RV and discount rate i , then we can easily calculate the value of the bond. Now suppose that an investor already knows the value of a bond in terms of its price, say, P and he is interested in finding a discount rate i , such that the above equation is satisfied, then this discount rate is called Yield to Maturity (YTM). So, YTM means the return on a bond (or discount rate) which makes the present value of all the coupons and redemption value equal to the price of bond P provided that the bond is held till its maturity. This is the return on investment which investors talk about when they invest in bonds. Thus, YTM means that i which satisfies the following equation:

$$P = \left[\sum_{k=1}^n C_k \times \frac{1}{(1+YTM)^k} \right] + \frac{RV}{(1+YTM)^n}$$

Although we have a **trial and error approach** (a very time-consuming approach) to calculate YTM, we have an approximation formula given below to calculate YTM if coupons are constant.

$$\text{Approximate YTM} = \frac{C + (F - P)/n}{(F + P)/2}$$

Where C = Constant coupon, F = Face value, P = Price of the bond

Example 12.3: Suppose that there is a bond with a face value of ₹ 1,000 and a coupon rate of 10%. If the maturity is 5 years and the price of the bond is ₹ 800, then calculate the approximate YTM. Interpret your result.

Solution:

$$\begin{aligned}\text{Approximate YTM} &= \frac{C + (F - P)/n}{(F + P)/2} \\ &= \frac{100 + (1000 - 800)/5}{(1000 + 800)/2} \\ &= \frac{100 + 40}{900} \\ &= \frac{140}{900} \\ &= 0.1555 \text{ or } 15.55\%\end{aligned}$$

Interpretation: It means that if ₹ 800 is invested in the bond held for 5 years, then it will give us a return of 15.55%.

12.5.2 Yield to Call (YTC)

In the case of a callable bond, the issuer can call or buy back the bond from the bondholder before the scheduled maturity date at a specified call price. Thus, redemption can be done before maturity. In such a case, the rate of return on the bond till the call date is called Yield to Call. Thus, YTC is that i which satisfies the following equation:

$$P = \left[\sum_{k=1}^s C_k \times \frac{1}{(1 + \text{YTC})^k} \right] + \frac{\text{Call price}}{(1 + \text{YTC})^s}$$

where,

s = Period up to call

12.5.3 Current Yield

The current yield is the annual interest payment divided by the current price of the bond. Thus, the current yield formula is

$$\frac{\text{Annual interest or coupon}}{\text{Current price of bond}}$$

Therefore, if a company's bonds with a 10% coupon were currently selling at ₹ 980, the bond's current yield would be 10.20% (₹100/₹980). It means by investing ₹ 1 now, an investor will earn ₹ 0.102 as interest at the end of 1

year. Current yield focuses on interest only not the redemption value and this is the reason we say that current yield is not an appropriate measure of yield.

Check Your Progress 3

- 1) What is **Yield to Maturity (YTM)**?
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- 2) Calculate the yield to maturity from the following information. Coupon rate = 7%, face value = ₹ 500, price of bond = ₹ 350, maturity period = 10 years. (**Hint: Use the approximation formula**).
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- 3) What are Yield to Call and Current Yield?
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12.6 CREDIT RATING

Credit rating is an evaluation of the creditworthiness of a business. It is a measure of the ability of the borrower to repay debt on time and in full. The agency performing credit rating is known as **a credit rating agency**. Credit rating agencies, assign credit ratings based on the borrower’s financial history, including factors such as income, assets, liabilities, repayment track record, and other relevant financial indicators. The credit rating agencies use a letter-based rating system to indicate the creditworthiness of borrowers, with AAA being the highest rating and D being the lowest. Credit ratings are important because they help lenders and investors assess the risk associated with lending money to a borrower. A high credit rating means that the borrower is less likely to default on their loans, which can lead to lower interest rates and better loan terms. On the other hand, a low credit rating can